BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 052 · Variables, Assignment and Data Types

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the difference between input and output?

Q2. Why is SSD random access often faster than HDD access?
A. Its read head spins faster
B. It uses a laser
C. Its platters are smaller
D. No read head must move

Q3. “Portability and durability mean exactly the same thing.” Correct this. Explain why.

Q4. A journey planner needs stops and times, but not bus-seat colours. “Abstraction removes every detail, even inputs.” Correct this. What should the planner keep?

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I Do · Read new knowledge

Key vocabulary

Variable — A variable stores a named value that can change while the program runs.

Assignment — The equals sign stores the new value in a variable, replacing its old value.

Integer — A whole-number data type is suitable for counts, ages and scores.

String — A string stores text such as names or class codes.

Algorithm — An ordered set of instructions to solve a problem.

Trace — Following instructions and recording changes to values or output.

New learning

What is Variable?: Variable — A variable stores a named value that can change while the program runs.

In this example: A library stock record uses assignment; The equals sign stores the new value in a variable, replacing its old value.

Follow the effect: A whole-number data type is suitable for counts, ages and scores. Explain how this changes the eventual program result.

Check understanding: String is used for a specific reason: A string stores text such as names or class codes.

Worked example

A library stock record is being written. Explain how variable and assignment are used to make the program work. [4 marks]

1. We are designing a library stock record. Identify which data and decisions are essential.

2. Variable [1] — A variable stores a named value that can change while the program runs. [1].

3. Assignment [1] — The equals sign stores the new value in a variable, replacing its old value. [1].

4. Two linked points for each technique give four mark points.

We Do · Practise with guidance

Guided 1

In a pupil merit totals, explain how variable helps the program. [2 marks]

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Worked method: Variable [1] — A variable stores a named value that can change while the program runs. [1].

Guided 2

For a sports-club members, explain why assignment matters. [2 marks]

Worked method: Assignment [1] — The equals sign stores the new value in a variable, replacing its old value. [1].

Guided 3

When designing a cinema seats remaining, explain how integer and string should be used. [4 marks]

Worked method: Integer [1] — A whole-number data type is suitable for counts, ages and scores. [1]. String [1] — A string stores text such as names or class codes. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Use the named constant to update the free RAM.

```
const RESERVED_MB = 120
freeMB = 500
freeMB = ...
print(freeMB)
```

write

Question 1

State two programming terms or techniques relevant to this lesson. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

A school register needs to work correctly. Explain how assignment and integer would help. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A pupil is programming a shop loyalty points. Explain the roles of variable, integer and string in their solution. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A variable stores a named value that can change while the program runs.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Input enters a program; output is information the program produces.

Q2. D — No read head must move

Q3. No moving read/write parts makes an SSD less likely to be damaged by shocks. Size and weight also affect portability.

Q4. Remove details that are not needed. Keep the information and parts needed to solve the problem, then link clear steps.

I Do / We Do pages · Guided practice

Q1. Variable [1] — A variable stores a named value that can change while the program runs. [1].

Q2. Assignment [1] — The equals sign stores the new value in a variable, replacing its old value. [1].

Q3. Integer [1] — A whole-number data type is suitable for counts, ages and scores. [1]. String [1] — A string stores text such as names or class codes. [1].

You Do pages · Code First

```
const RESERVED_MB = 120
freeMB = 500
freeMB = freeMB - RESERVED_MB
print(freeMB)
```

Expected result: 380

You Do pages · Independent questions

Q1. Variable [1]; Integer [1].

Q2. Assignment [1] — The equals sign stores the new value in a variable, replacing its old value. [1]. Integer [1] — A whole-number data type is suitable for counts, ages and scores. [1].

Q3. Variable [1] — A variable stores a named value that can change while the program runs. [1]. Integer [1] — A whole-number data type is suitable for counts,

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

ages and scores. [1]. String [1] — A string stores text such as names or class codes. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: